

■ Interview-Ready Sheets

■ Sheet 1: Core CS Subjects

■ Goal → Be clear in fundamentals, repeatedly asked in interviews.

1. DBMS

- ER diagrams, Normalization (up to BCNF)
 - SQL queries (Joins, Subqueries, Group By, Having)
 - Indexing, Transactions (ACID)
 - Deadlock, Isolation Levels
- Importance → Very high in service + product interviews.

2. OOPS (Java)

- Classes, Objects, Constructors
 - Inheritance, Polymorphism (Runtime vs Compile-time)
 - Abstraction, Interfaces
 - Encapsulation, Access Modifiers
 - Static, Final, this, super
 - Overloading vs Overriding
- Importance → Always asked, often through design questions.

3. Operating Systems

- Processes vs Threads
 - Scheduling Algorithms
 - Deadlocks (Banker's Algo)
 - Memory Mgmt (Paging, Segmentation, Virtual Memory)
 - Semaphores, Monitors
- Importance → Medium to high, esp. product-based.

4. Computer Networks

- OSI vs TCP/IP layers
 - IP, TCP vs UDP
 - DNS, DHCP, HTTP/HTTPS
 - Routing basics
- Importance → Moderate, often 1-2 questions.

5. Misc

- Compiler Basics (Lexer, Parser)
- Basics of Cloud (AWS/Azure fundamentals)

■ Sheet 2: System Design (HLD + LLD)

■ Goal → Show awareness of scalability, design choices.

High-Level Design (HLD)

- Load Balancers, CDN
 - Horizontal vs Vertical scaling
 - Databases (SQL vs NoSQL)
 - Caching (Redis, Memcached)
 - Message Queues (Kafka, RabbitMQ)
 - CAP Theorem
- Impact → Asked in senior roles, but for freshers: explain in project context.

Low-Level Design (LLD)

- Class diagrams (UML basics)

- Designing small systems (Parking Lot, Library)
- LRU Cache, Rate Limiter
- Solid Principles (SRP, OCP)
- Impact → Good for product-based interviews, shows maturity.

■ Sheet 3: Development Stack (Next.js, React, PostgreSQL, APIs)

- Goal → Build and defend projects confidently.

1. React

- Components, Props, State
- Hooks (useState, useEffect, useRef, useContext)
- Event Handling, Conditional Rendering
- Forms, Lists & Keys
- Routing basics

2. Next.js

- File-based Routing
- API Routes (/api/*)
- SSR vs SSG vs CSR
- Data Fetching (getServerSideProps, getStaticProps)
- Middleware for auth
- Deployment (Vercel)

3. PostgreSQL

- Schema design, Constraints
- Joins, CRUD
- Indexes, Transactions

4. Backend/API Basics

- REST APIs (GET, POST, PUT, DELETE)
- Auth (JWT, Sessions)
- Error handling, Middleware
- Security: Hashing, Input sanitization

5. Extra

- State Management (Redux/Context)
- WebSockets (real-time)
- Caching (Redis)
- Testing (Jest)